Lecture 1 - Introduction

Causal Inference and Course Info.

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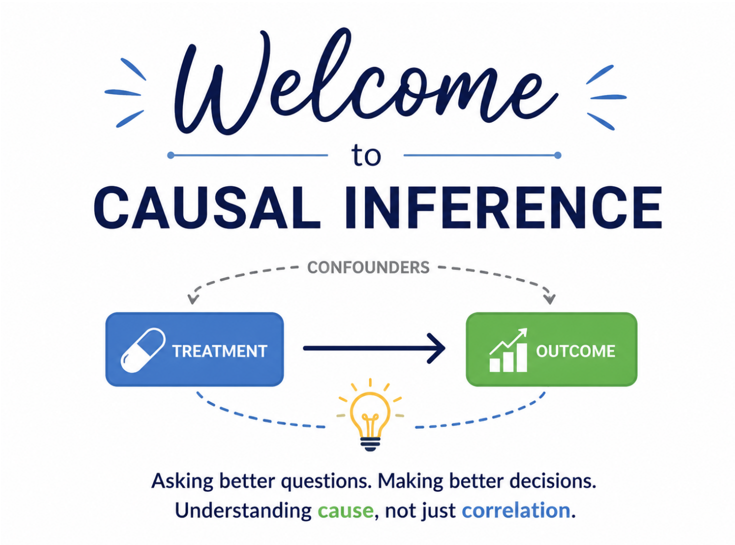
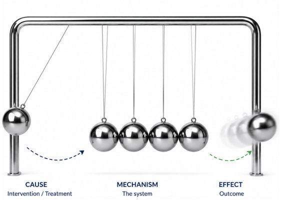
Welcome!


Welcome
to
CAUSAL INFERENCE
CONFOUNDERS
TREATMENT
OUTCOME
Asking better questions. Making better decisions.
Understanding cause, not just correlation.

Self-introduction



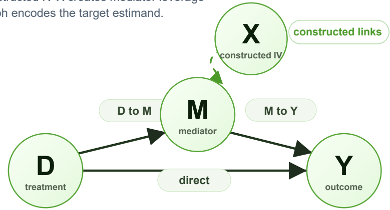
Xi Chen

- Developing and applying methods and insights of causal inference for:
    - Digital marketing
    - Policy evaluation
    - Machine learning
    - Behavioral research

Constructed instrument for mediation analysis

Causal graph + constructed instrument

Constructed IV X creates mediator leverage
Graph encodes the target estimand.



weaker assumptions | diagnostics | ACME / ADE inference

Tool-building workflow

From identification to reusable software.

1 graph | 2 C-IV | 3 bootstrap

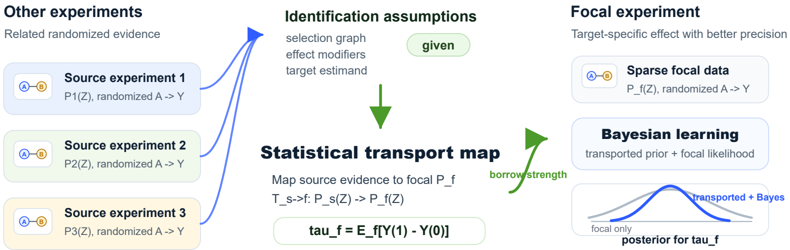
```
library(bootmed)
causal_med_civ(data,
x = "D", m = "M", y = "Y",
iv = construct_IV(...),
B = 2000)
```

Output

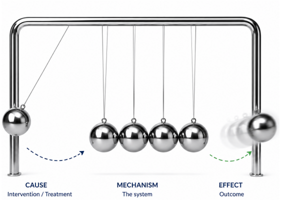
IV strength	$\chi^2 = 432.8$	$p < .001$
ACME	bootstrap 95% CI	

interpretable indirect/direct effect estimates

Learning treatment effect across experiments



Self-introduction



A self-introduction with:

- Your name and background.
- What to expect from the course.

21 Questions Game: causal inference version

- No causation without correlation.
- Causal inference methods are more credible.

- Running a regression gives you effects:
  e.g., Sales $= b \times$ Price
- Adding more variables always gives you better estimates:
  e.g., Sales $= b \times$ Price $+$ Seasonality $+ \cdots$

21 Questions Game: causal inference version

"No causation without manipulation."

- "Only experiments can establish causality."
- "Causes are only those things that could, in principle, be treatments in experiment" (Holland 1986, p. 954).
- "Such questions ['no series of actions can be inferred from the description of the treatment'] have no causal answer within our framework" (Rubin 1978, p. 39).

Questions

- How do experiments establish causation?
- What is so unique about experiments?

21 Questions Game: causal inference version

A or B?

A. Causation can be inferred solely from data.
B. Causal inference always requires assumptions on top of data.

Any other questions?

1. The Opening
2. On Causal Inference
3. Course Overview
4. References

What is causal inference?

Inferring the effects of any treatment / policy / intervention / business strategy, etc.

Examples

Think about the research questions that you are interested in...

Why causal inference?

Correlation $\neq$ Causation.

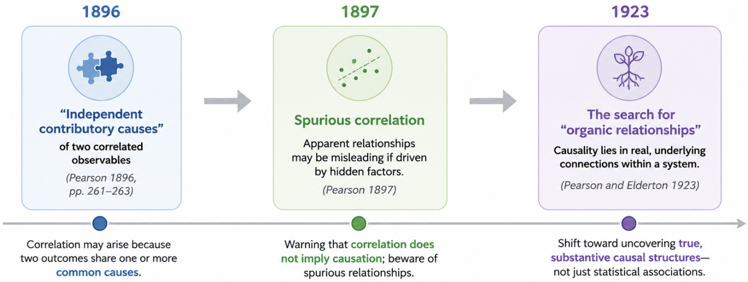
Correlation is not causation

This is the idea from Karl Pearson – the inventor of correlation!

> *"If the unit A be always preceded, accompanied or followed by B, and without A, B does not take place, then we are accustomed to speak of a causal relationship between A and B."*
>
> ***Pearson and Lee (1897, p. 459)***

> *"It is the conception of correlation between two occurrences embracing all relationship from absolute independence to complete dependence, which is the wider category by which we have to replace the old idea of causation."*
>
> ***Pearson (1910, p. 157)***

The origin of "correlation is not causation"



"Beyond such discarded fundamentals as "matter" and "force" lies yet another fetish amidst the ***inscrutable arcana of modern science****, namely, the category of cause and effect."*

Karl Pearson (1892)

The importance of knowing causality: "cliché" examples

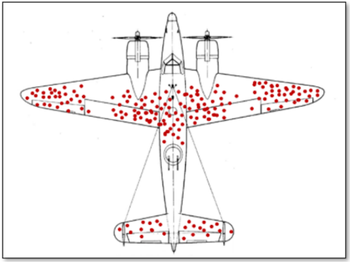


Figure 1: Survival Bias (Wald 1943)

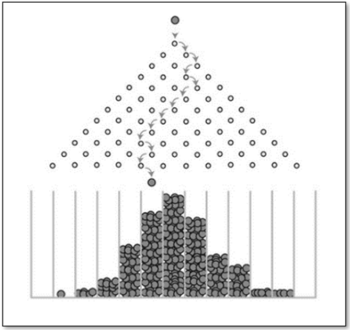


Figure 2: Regression to the Mean (Galton 1894)

The importance of knowing causality

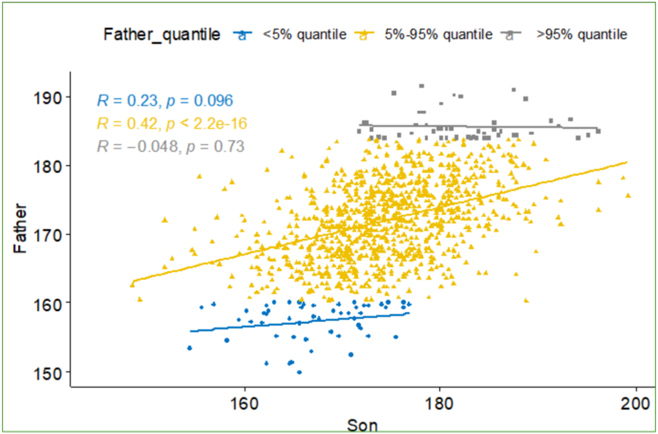


Figure 3: Regression to the Mean (Galton 1894)

The importance of knowing causality

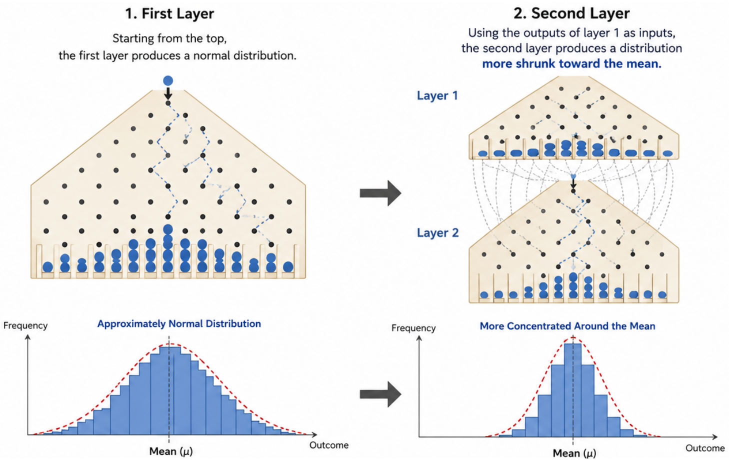
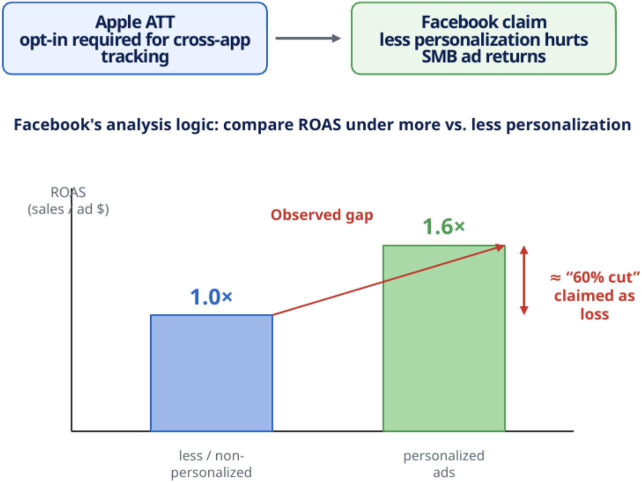


Figure 4: Two-layered Galton Board

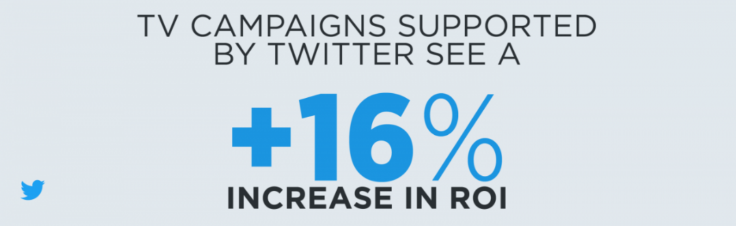
Business examples

Facebook's campaign against Apple's privacy policy[1].



[1] For more, see: `https://hbr.org/2021/02/`

Business examples



"..., we looked at nearly four years of paid theatrical history data - from December 2011 to June 2015. The research team used ***multivariate regression analysis*** *(a process that measures and predicts the sales* ***impact*** *of various media channels) to understand the* ***effects*** *of changes in Twitter media for movies"*[2].

[2]Source: https://blog.twitter.com/en_us/a/2016/new-movie-marketing-research-reveals-twitter-ads-deliver-increased-tic

Flops in scientific research

What makes causal claims more credible?

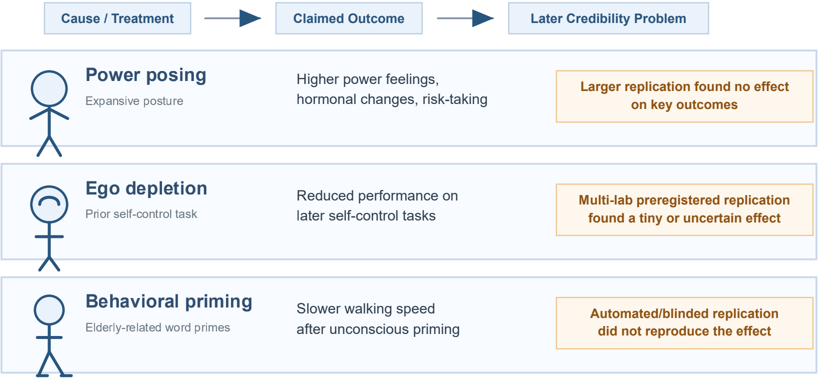


Figure 5: Examples in Behavioral Research

Flops in scientific research

What makes causal claims more credible?

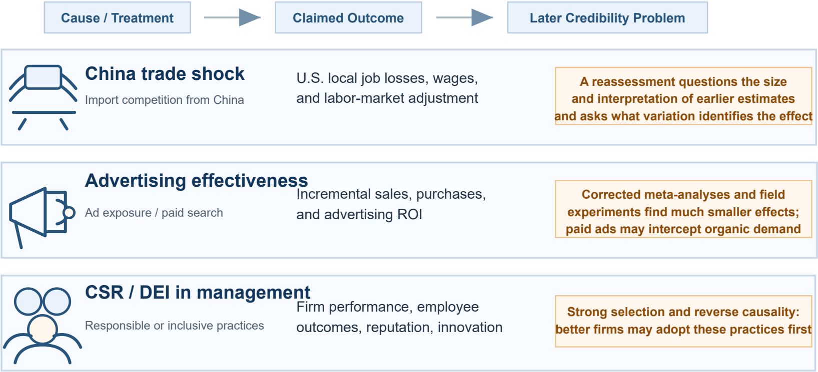


Figure 6: Examples in Econ and Business Research

Causal Claim Clinic

The Clinic Template

1. What is the causal claim?
2. What is the treatment or cause variable?
3. What is the outcome variable?
4. What is the comparison?
5. Why the claim fail?
6. What would make the claim more credible?

Causal Claim Clinic

The Clinic Template

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Key question

What makes a causal claim credible?

The thinking behind the course design

The decision problem:

$$\max \quad \text{LearningExperiences}$$
$$\text{subject to:} \quad \begin{cases} \text{5 days and 25 hours} \\ \text{Compact Learning} \\ \text{A Complex Topic} \\ \text{Diversity of Our Group} \end{cases}$$

The thinking behind the course design

To integrate different perspectives for problem-solving

To focus on the in-depth understanding

Only on essential technical details

To communicate "principles" of causal inference

For your critical thinking

To use blended learning methods

To address class diversity

To gain hands-on experiences

To reduce your prep efforts

Foundation of the course

A paradox in the empirical adoption of causal inference methods.

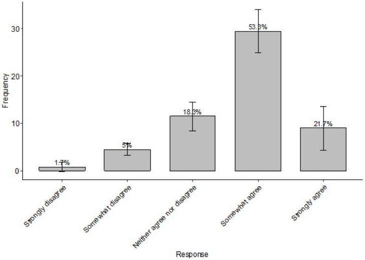


Figure 7: Causal inference methods are generally more credible.

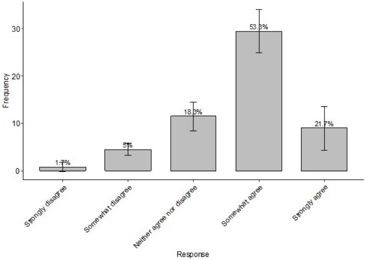
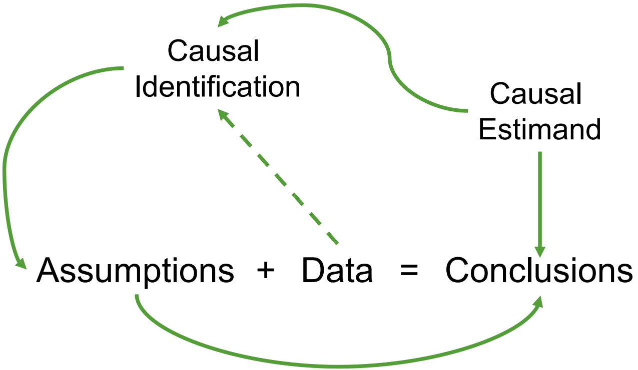


Figure 8: The credibility of causal inference applications varies significantly.

Foundation of the course

The fundamental question in causal inference:

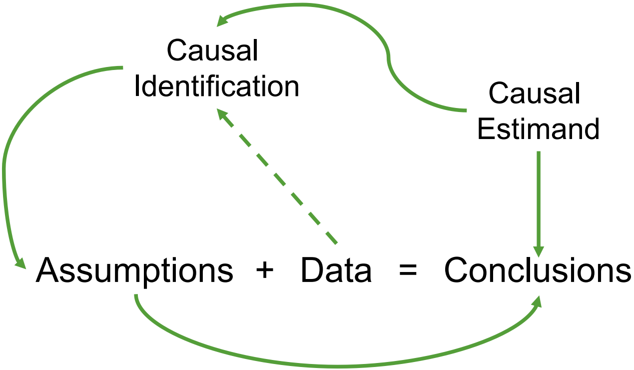
What makes causal inference methods credible?



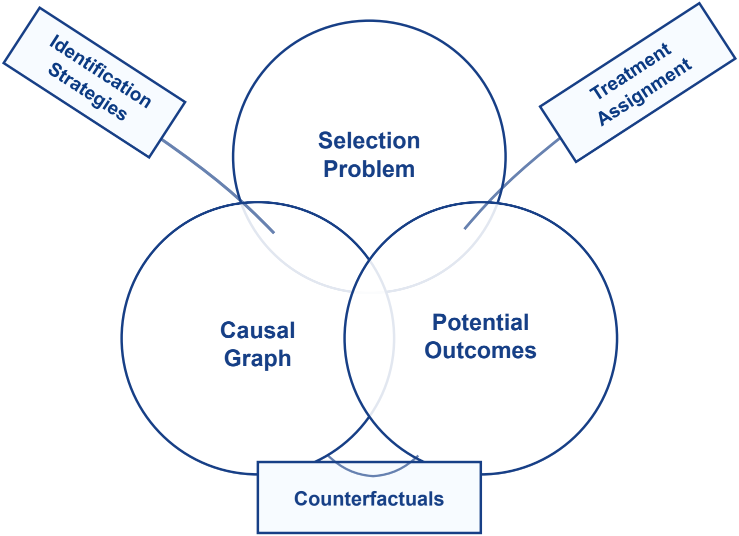
Foundation of the course

What makes causal inference methods credible?

- Assumptions are not unique.
- Different assumptions $\rightarrow$ different conclusions.
- Credibility of assumptions $\rightarrow$ credibility of conclusions.



A multi-disciplinary perspective



Design logic of the course

Topics	Controls
Completely randomized experiments	Full control Full information
Matching and weighting	Partial control Full information
Non-compliance in experiments	Partial control Partial information
Quasi-experiments	No control Partial or no information

Presentation of each topic

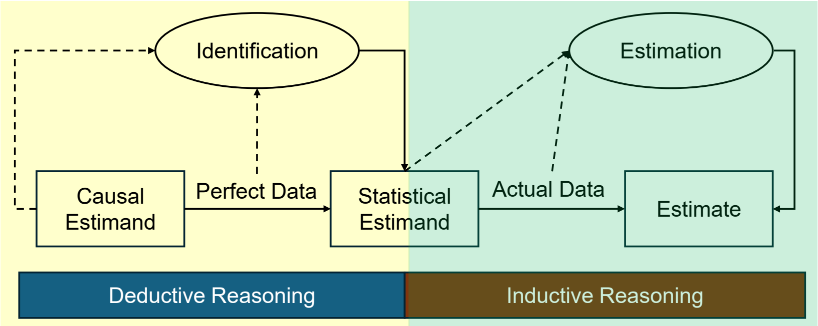


Figure 9: The identification-estimation flow

Instruction methods

Lectures: principles of causal inference

- Basic concepts and understanding.
- Systematic way of implementation.
- Limitations and boundaries of tools.

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Discussion: learning through questioning

- Discussing selected topics based on papers or cases.

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Cases

- Apply models to solve real problems.
- Data- and model-driven.
- Step-by-step guide for doing projects.

Course materials

Textbooks / monograph
(optional; for references)

Assumptions in Causal Inference: Illuminating the Path to Credibility

Suggested Citation: Xi Chen (XX), "Assumptions in Causal Inference: Illuminating the Path to Credibility", : Vol. xx, No. xx, pp 1–XX. DOI: XXXXXXXXX.

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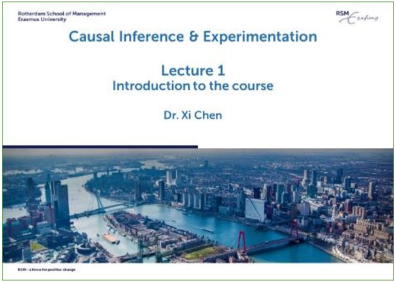
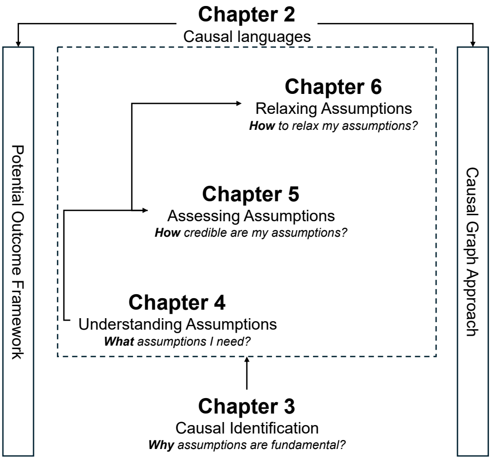
Slides



Basic logic of the monograph

The presentation of the course follows closely the logic of the monograph.



Course materials

Papers for discussion

- On GitHub
- Required
- Try to skim through them

Extended readings

- Optional
- If you are interested, read at your leisure time

Assessment

Integrative assignment

- Three weeks to complete.
- Online on the weekend; data and questions on GitHub.
- Data analytics for causal inference problems.
- Two types of questions:
  - Analytical and interpretation questions.
  - Open conceptual or managerial questions.
- Focus on problem-solving.

Any other questions?

Any other questions?

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